

XAI-Driven Clinical Decision Support System for Interpretable Anemia Classification Using Raw Hematological Parameters



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Introduction

Anemia is a major global health problem caused by low hemoglobin or insufficient red blood cells. Early detection is important to prevent severe health complications. Previous studies applied machine learning models such as Decision Tree, Ensemble Voting, and KNN for anemia prediction; however, most **lacked explainability** and **real-time clinical interpretability** [1]. This research proposes an Explainable AI-based anemia classification framework using hematological parameters, **Random Forest**, and **SHAP explainability** to provide accurate, transparent, and clinically interpretable predictions.

Problem Statement & Research Objectives

Problem Statement:

Manual anemia as black-box systems without clinical interpretability. [1]. Explainable diagnosis may cause delayed or inconsistent clinical decisions. Existing AI models often operate and accurate AI systems are needed for trustworthy anemia prediction.

Objectives:

- Develop an AI-based anemia classification system using hematological parameters
- Build and evaluate machine learning models for **accurate anemia prediction**.
- Integrate **SHAP-based explainability** for interpretable prediction analysis.
- Develop a real-time **clinical decision support** application for anemia screening.

Dataset Overview

Source	Medical Centre Hospital, Chattogram, Bangladesh [2]
Records	1004 patients
Features	9 hematological parameters
Classes	2 (Anemia / Normal)

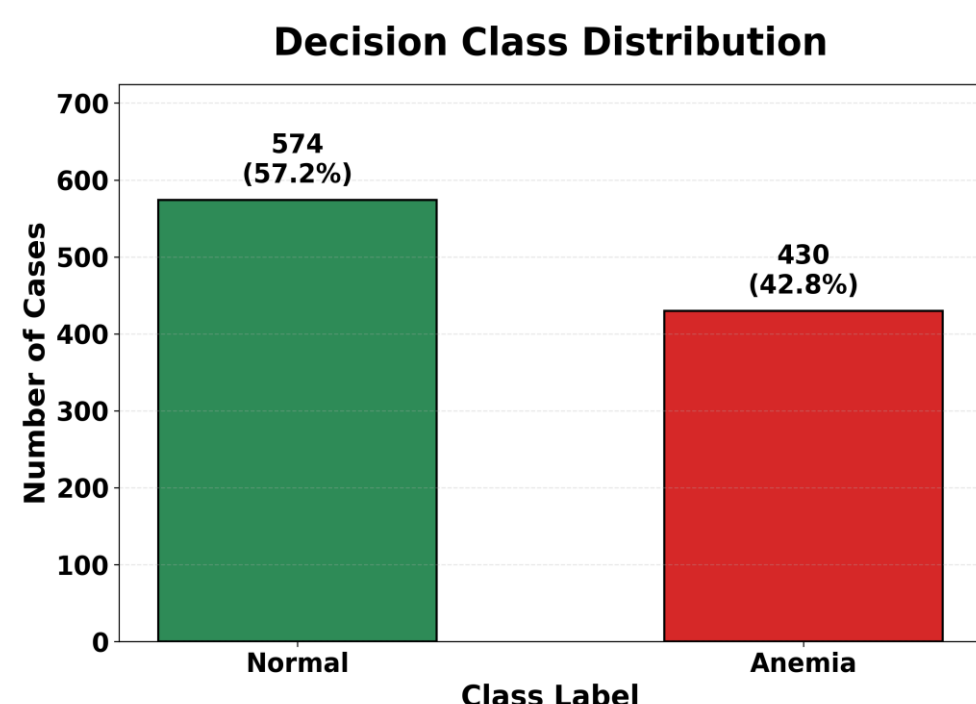


Figure 1: Distribution of the target classes in the hematological dataset

Methodology

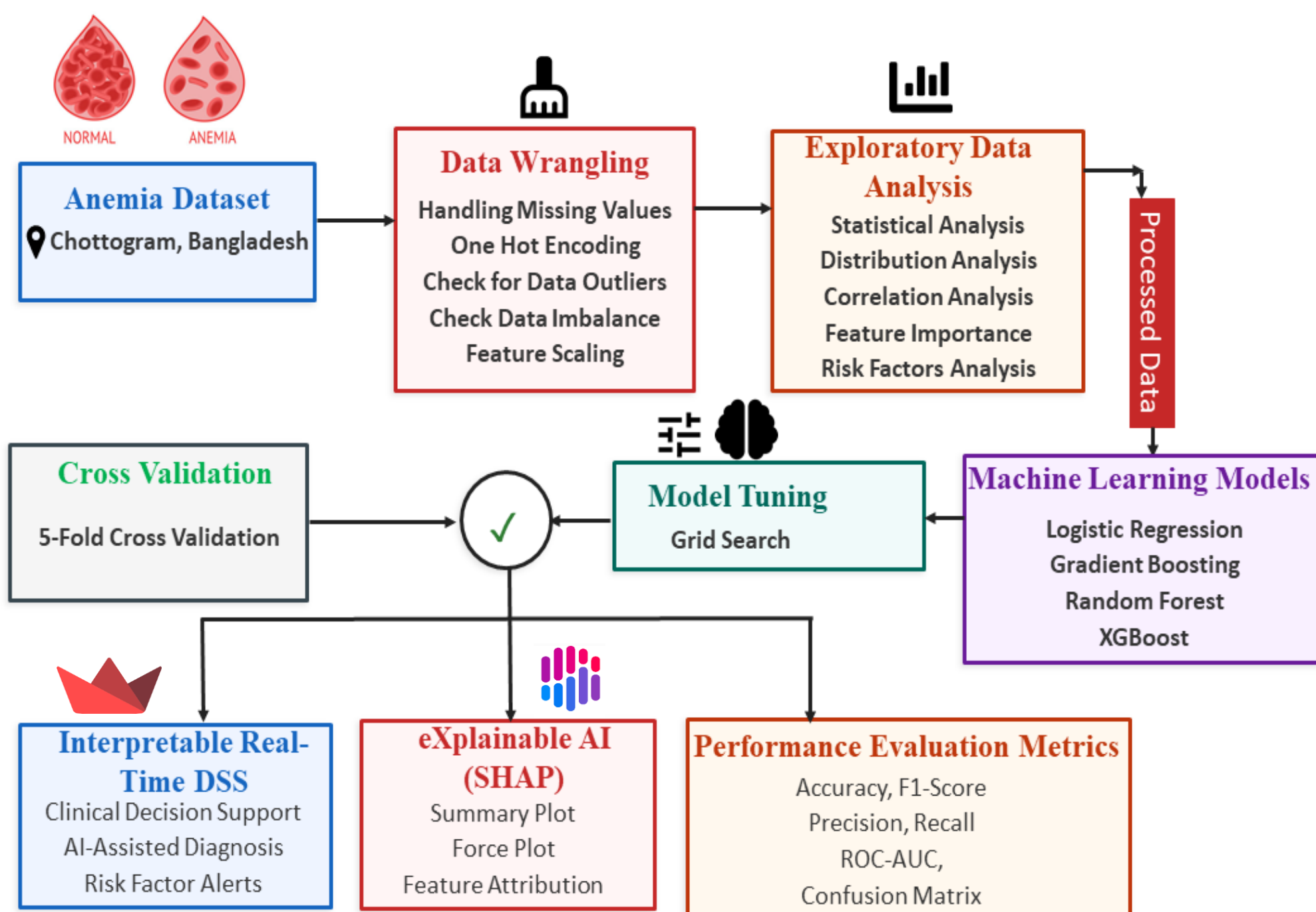


Figure 2 : Overall workflow of the proposed methodology for anemia classification.

Key Contributions & Novelty:

- **Achieved 98.98% accuracy** using Random Forest and introduced a Hb-MCV Ratio feature engineering.
- **Integrated SHAP** for explainable and interpretable AI predictions.
- Developed a **real-time clinical decision support** system for anemia classification.

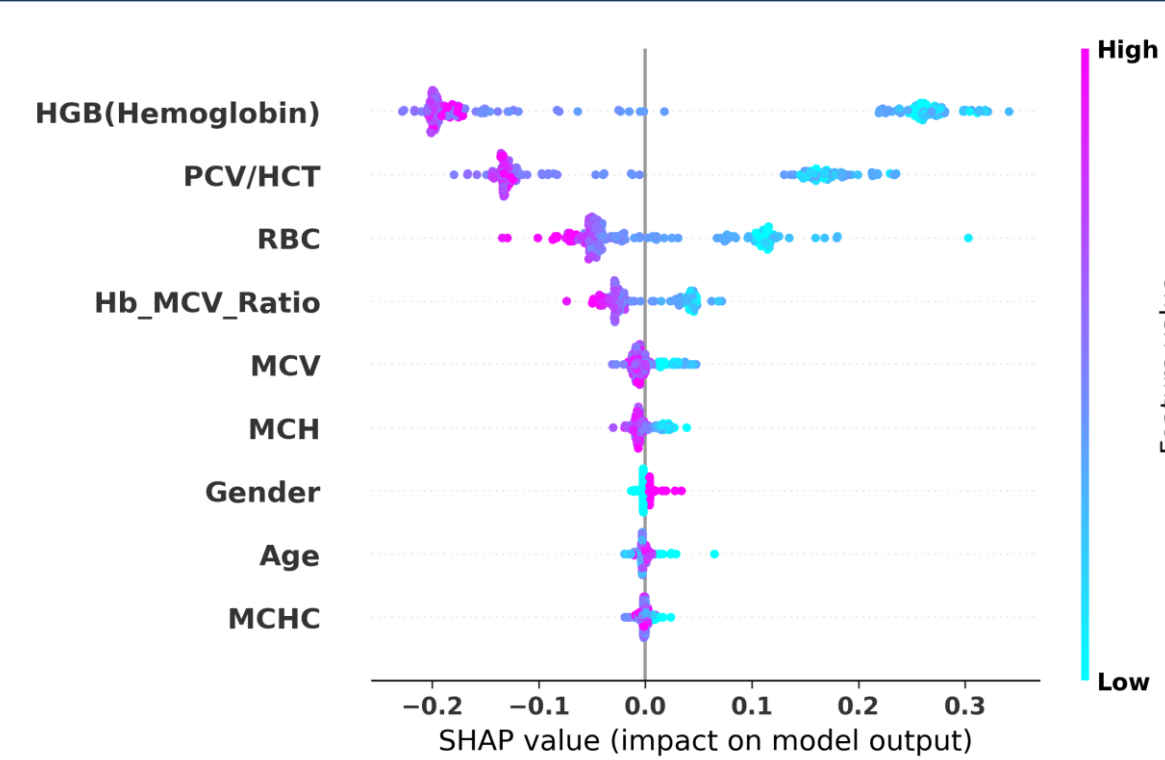
Key Findings & Results

Best predictive model Performance Comparison

Model	Accuracy
Logistic Regression	97.50%
Gradient Boosting	98.80%
XGBoost	97.50 %
Random Forest ★	98.98%

Proposed Model Comparison With SOTA Performance

Reference	Best Model	Accuracy
Gómez et al. (2024)	Decision Tree	98.30%
Darshan et al. (2024)	KNN Classifier	86.33%
Hasan et al. (2023)	Ensemble Voting	96.67%
Proposed Method	Random Forest	98.98%



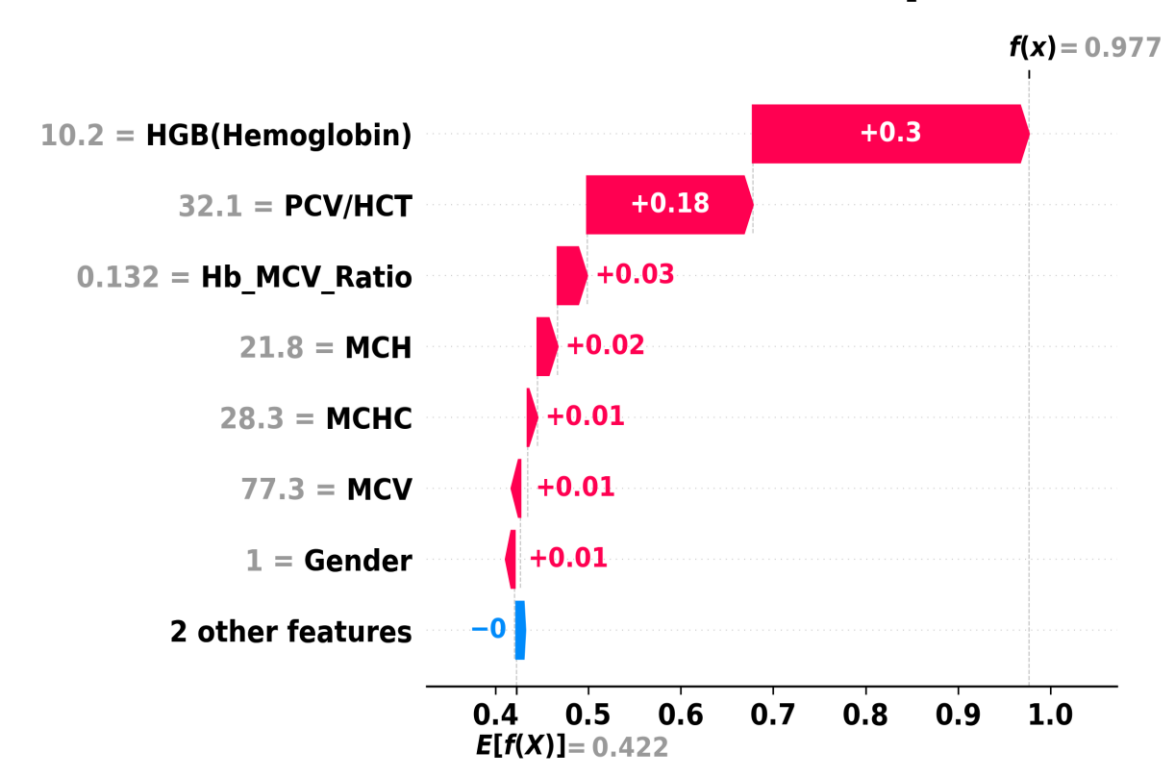
SHAP Value Impact Analysis

Confusion Matrix of Random Forest Model

True Label \ Predicted Label	Normal	Anemia
Normal	114	0
Anemia	2	81

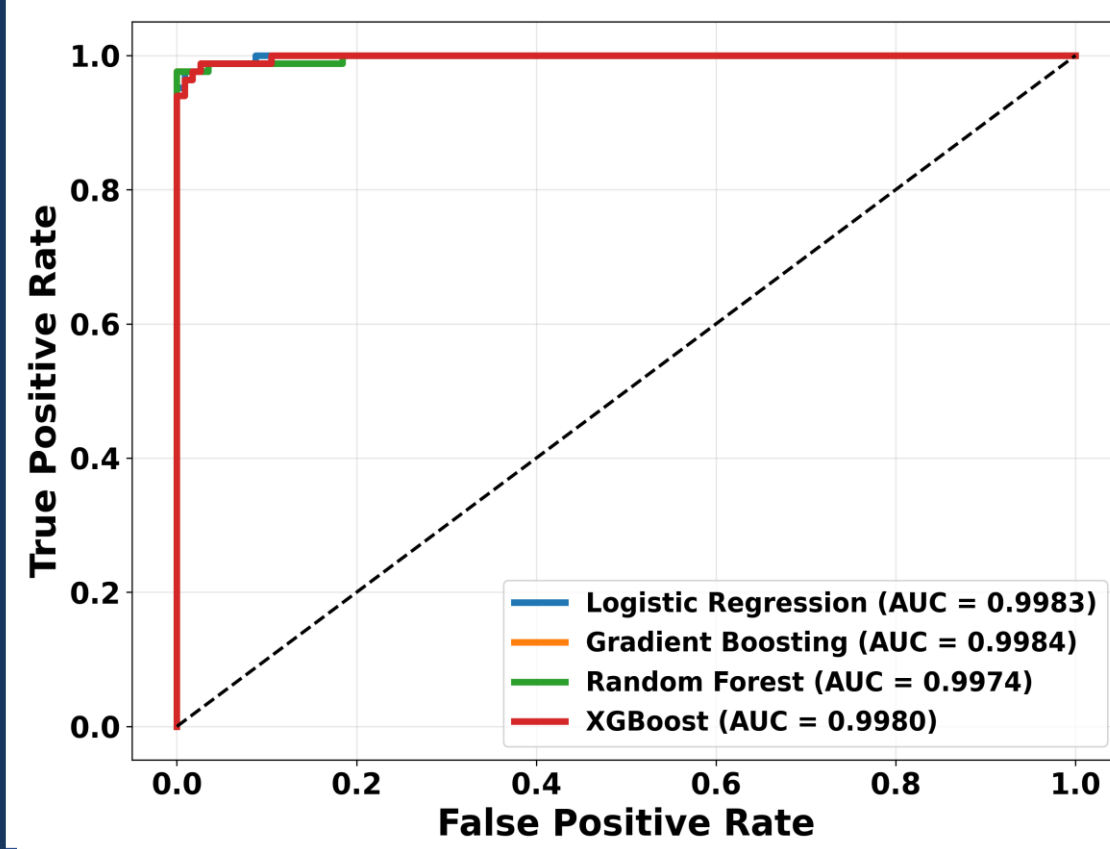
Random Forest Classification Performance

SHAP Waterfall Explanation



Shape Value Waterfall Explanation

ROC-AUC Curve



ROC-AUC Performance Comparison

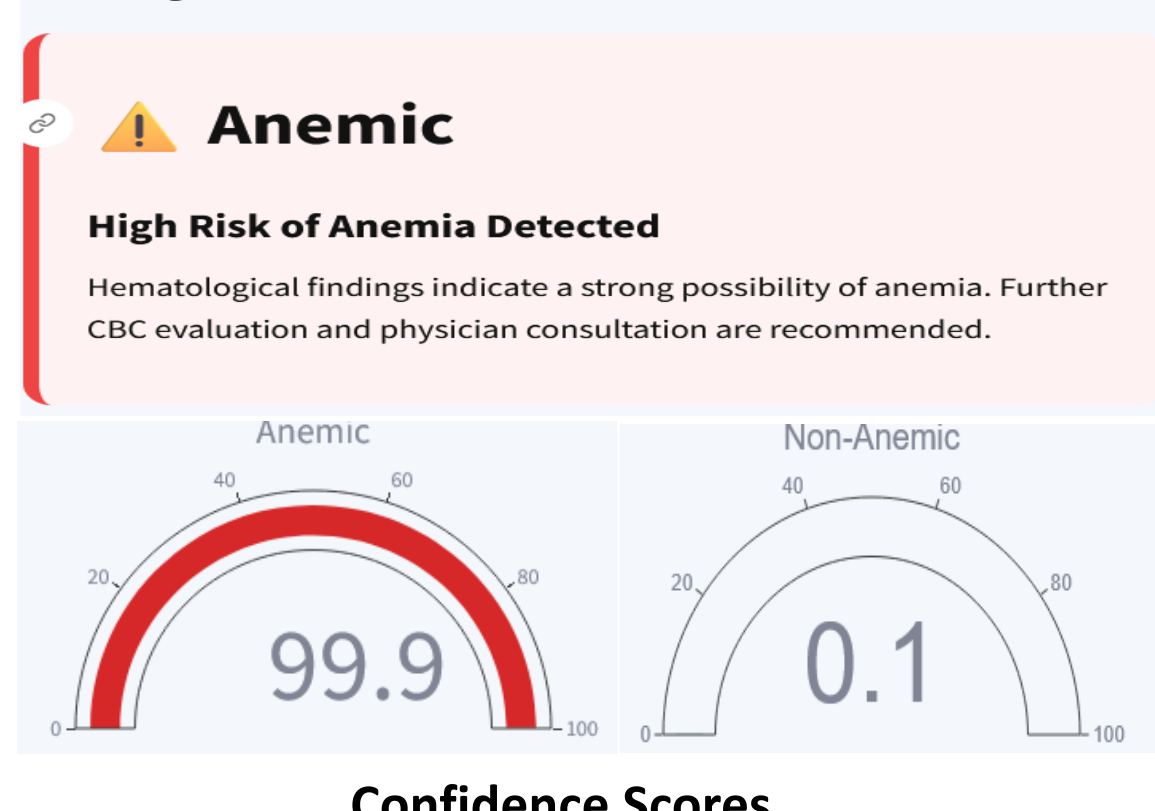
Explainable AI for Anemia Classification

Clinical Decision Support System using Random Forest and SHAP Explainability

Patient Hematological Parameters

Age (years)	80	PCV/HCT (%)	25.20
Gender	Female	Normal Range: 36-54%	
Hemoglobin (g/dL)	8.30	MCV (fL)	82.90
Normal Range: 12-18 g/dL		Normal Range: 80-100 fL	
RBC (10 ¹² /μL)	3.04	MCH (pg)	27.30
Normal Range: 3.9-4.1 × 10 ¹² /μL		Normal Range: 26-34 pg	
		MCHC (g/dL)	32.90
		Normal Range: 31-36 g/dL	

Analysis Results



Conclusion & Clinical Impact

Our proposed Explainable AI-based Clinical Decision Support System provides a **reliable, rapid, and interpretable** approach for automated anemia classification using hematological biomarkers. The integration of SHAP-based Explainable AI enhances clinical transparency by identifying key factors influencing predictions, thereby **improving clinician trust and supporting** informed medical decision-making.

Business & Entrepreneurial Impact:

- **Startup Potential:** Scalable B2B SaaS platform for automated screening in labs and clinics.
- **Revenue Model:** API-based subscriptions for hospitals and freemium tiers for practitioners.
- **Zero Hardware Cost:** Operates on existing raw CBC data without new equipment.
- **Social Entrepreneurship:** Democratizes expert-level screening in remote areas (rural healthcare).

Future Work:

- Integrate **advanced deep learning and ensemble** models for improved prediction accuracy
- Expand the framework using larger **real-world multicenter clinical datasets**
- Develop a **cloud-based** real-time clinical decision support application with enhanced Explainable AI features

References

- [1] M. M. Rahman, M. U. Mojumdar, H. A. Shifa, N. R. Chakraborty, N. P. Stenin, and M. A. Hasan, "Anemia Disease Prediction using Machine Learning Techniques and Performance Analysis," in *Proc. 11th Int. Conf. Computing for Sustainable Global Development (INDIACom)*, New Delhi, India, 2024, pp. 1276–1282, doi: 10.23919/INDIACom61295.2024.10498962.
- [2] M. S. Khan, S. T. Shovan, H. A. Shaikat, M. M. Ali, N. Parvez, and M. Assaduzzaman, "Raw Hematological Dataset for Anemia Analysis and Classification," *Mendeley Data*, vol. 1, 2026, doi: 10.17632/tnswkb4bt8.1.

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